

UNIT 5 · TOPIC 5.4

Blockchain in Air Ticketing

A shared, tamper-proof record for travel.

B.B.A. Aviation Management · Semester III · Unit 5: Technology & Trends
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TOPIC 5.4

Mission Video Briefing



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02

TOPIC 5.4

Mission Question



THINK ABOUT IT

Could an air ticket be made impossible to fake? How?

**THINK**

30 seconds alone

**PAIR**

1 minute with a neighbour

**SHARE**

2-3 answers with the class



TOPIC 5.4

Mission Briefing

1

5.4.1

Blockchain definition and applications in the airline industry

2

5.4.2

How blockchain enhances security, transparency and efficiency

3

5.4.3

Real-world examples and case studies

4

5.4.4

Future potential in reducing fraud and improving passenger experience



MISSION OBJECTIVES

- Explain blockchain in simple terms
- Identify airline use cases
- Review real pilots
- Evaluate future potential and limits

Pedagogy: Technology-Based Learning

TOPIC 5.4.1

Blockchain in Plain Words

**Chain of blocks**

Records grouped in blocks, each linked to the previous one.

**Shared ledger**

Many parties hold the same copy.

**Tamper-evident**

Changing past records is practically impossible.

**ANALOGY**

A class attendance register that every student holds a copy of. If one person changes their copy, it no longer matches everyone else's — so the change is rejected.



TOPIC 5.4.1

Airline Applications

**Tickets & Orders**

Tokenised tickets that can't be duplicated.

**Loyalty**

Points as digital tokens, easy to spend.

**Settlement**

Faster interline and agent settlement.

**Aircraft Parts**

Trusted maintenance records.



TOPIC 5.4.2

Security, Transparency, Efficiency



Security

Cryptography protects records from fraud.



Transparency

All parties see the same data.

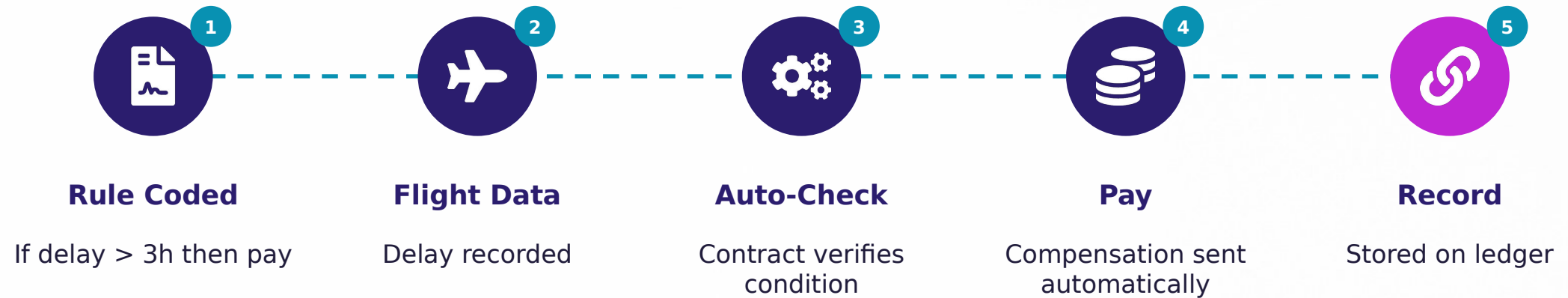


Efficiency

Smart contracts auto-execute payments/refunds.

TOPIC 5.4.2

Smart Contract Refund (Concept)



TOPIC 5.4.3

Real-World Examples

Example	What it did
Singapore Airlines KrisPay (2018)	Blockchain-based loyalty wallet to spend miles at partners
Winding Tree	Open-source travel distribution platform tested with airlines
IATA / SITA pilots	Industry trials in settlement, identity and data sharing
Parts traceability	MRO pilots for aircraft component histories

Note: Many projects remain pilots — adoption is gradual.

TOPIC 5.4.3

A Ticket as a Token (Concept)

How a blockchain ticket record might look.

- Each ticket is a unique token.
- Ownership and status on the ledger.
- Transfers only if airline rules allow.
- Duplicate tickets impossible.

LEDGER ENTRY (concept)

```
BLOCK    #10245
TOKEN     TKT-AI-7f3a9c
OWNER     0x8a1...b21 (verified ID)
FLIGHT    AI887 20NOV DEL-BOM
STATUS    OPEN
PREV      #10244  HASH 9e2c...
```

TOPIC 5.4.4

Potential vs Limits



Potential

- Fraud-proof tickets
- Instant settlement
- Self-sovereign digital ID
- Automatic compensation

VS



Limits

- Needs industry-wide agreement
- Privacy vs permanent records
- Integration cost
- Regulation still developing



TOPIC 5.4.4

Where Blockchain Stands

Pilots

most airline uses are trials

Loyalty

most mature use case so far

Future

identity and settlement show promise

TOPIC 5.4

Innovation Lab: Blockchain Classroom Ledger

**PEDAGOGY****Technology-Based Learning****TIME · 10 min****FORMAT · Whole class**

- 1 Five students each hold a copy of a “ticket ledger” sheet.
- 2 Record 3 ticket sales; everyone updates their copy.
- 3 One student secretly alters a record — class finds the mismatch.
- 4 Discuss how this prevents fraud.



TOPIC 5.4*Click to reveal each answer*

System Check Quiz

Q1**Blockchain records are...**

- A) Easy to edit
- B) Tamper-evident
- C) Paper-based
- D) Private to one person

ANSWER**B****Q2****KrisPay is linked to...**

- A) Emirates
- B) Singapore Airlines
- C) IndiGo
- D) Qantas

ANSWER**B****Q3****Code that auto-executes when conditions are met is a...**

- A) Smart contract
- B) PNR
- C) SSR
- D) BSP

ANSWER**A****Q4****Most airline blockchain uses today are...**

- A) Fully deployed everywhere
- B) Pilots/trials
- C) Banned
- D) Paper

ANSWER**B**

MISSION COMPLETE

Key Takeaways

- ✓ Blockchain = shared, tamper-evident ledger
- ✓ Uses: tickets, loyalty, settlement, parts
- ✓ Examples: KrisPay, Winding Tree, industry pilots
- ✓ Big potential, but adoption is gradual

EXIT TICKET Write 1 thing you learned today + 1 question you still have.

NEXT MISSION

**Topic 5.5 — AI and
Machine Learning in
Ticketing**

Thank you · Questions?